Memory masking versus overwriting in procedural categorization

Keywords: procedural learning; memory masking; stimulus-response associations; information-integration

Abstract

Behaviors acquired through procedural learning systems such as athletic skills are highly robust and notoriously resistant to change, even when they become contaminated by maladaptive habits. Identifying effective methods for modifying such procedural knowledge is therefore a major challenge. Previous work suggested that feedback contingency, that is, the degree to which behavior causes outcomes, is a critical factor determining whether procedural knowledge can be modified. Specifically, Crossley, Ashby, and Maddox (2013) reported that an intervention combining random and veridical feedback appeared to erase recently acquired procedural category knowledge. However, their study could not establish whether the intervention truly erased prior learning or simply suppressed its behavioral expression, leaving open the possibility it could reemerge under appropriate conditions. In the present study, we directly tested this possibility. Across two experiments, we examined whether this intervention leads to true overwriting of stimulus-response associations or behavioral masking. Decision-bound modeling and Bayesian estimation of reacquisition rates provided clear evidence that mixed random-veridical feedback does not erase procedural knowledge. Instead, learning rapidly re-emerged when participants were verbally cued that valid feedback had resumed. These findings necessitate a substantial revision of existing models of procedural category learning and have important implications for approaches to changing procedural skills more broadly.

Introduction

Habitual behavior has become increasingly prevalent in the modern environment, where devices and social media platforms are intentionally designed to promote repeated use [1, 2]. A growing body of evidence links such habits to negative outcomes, including attentional fragmentation, reduced productivity, and heightened risks for anxiety and depression [3, 4, 5, 6, 7, 8, 9]. Beyond individual consequences, the large-scale emergence of digital habits has been argued to shape patterns of public discourse, social cohesion, and even political polarization [1, 2]. Clarifying how such behaviors can be disrupted or regulated represents a pressing challenge for contemporary basic and applied research.

A major source of habitual behavior is likely to arise from the neural mechanisms underlying operant stimulus-response (SR) learning [10, 11, 12]. In the animal learning literature, especially in rodent models, behavior is often characterized as either habitual or goal-directed [10, 11]. Habitual behaviors are rigid, triggered directly by environmental cues, and often persist even when the associated reward is devalued. For example, a rat may continue pressing a lever for food despite being sated, much like a human may continue scrolling through social media absent any specific goal or gratification. By contrast, goal-directed actions are highly flexible and sensitive to the agent's current goals and motivational states. A goal-directed rat that has learned to press a lever for food will stop doing so if the food is devalued (e.g., by pre-feeding), just as a goal-directed human would stop scrolling through social media if they had just received gratification from it or had a more pressing task to attend.

The distinction between habitual and goal-directed behavior has also been made in the human literature [13], and closely parallels the division between procedural and declarative learning and memory systems [14, 10]. Procedural memory underlies behaviors learned through reinforcement of repeated experience, typically with

little cognitive effort or conscious awareness. By contrast, declarative memory supports flexible, hypothesis-driven reasoning and explicit knowledge. This division is particularly well studied in the domain of category learning, where different category structures are thought to preferentially engage different memory systems [15, 16, 17].

Crossley, Ashby and Maddox (2013) reported an intervention that appeared to induce true overwriting of procedural category knowledge [18]. Their study involved exposing participants to a mixture of random and veridical feedback following category learning. Participants who had previously acquired a category structure through procedural learning no longer showed evidence of this knowledge following the intervention, suggesting that the learned associations had been erased. However, rather than eliminating the procedural memory, it is possible that the intervention only masked its behavioral expression. Under this possibility, the underlying SR mappings remain intact but lie dormant and may reemerge under appropriate conditions.

In the present study, we directly test this possibility. We provide clear evidence that the intervention purported by Crossley et al. (2013) to overwrite existing procedural learning instead just masks the expression of that knowledge. These findings have important implications for how we understand, measure, and intervene on habitual behaviors across domains.

Methods

Experiments and conditions

The study consisted of two between-subjects experiments, each involving three phases: Learn, Intervention, and Test. Within each experiment, participants were randomly assigned to one of two between-subject conditions: *Relearn* or *New Learn*. The Learn and Test phases were identical across experiments and conditions. During the Learn phase, participants were trained on a category structure designed to promote procedural learning. In the Test phase, they were assessed either on the same category structure (Relearn) or on a new one (New Learn). The critical manipulation occurred during the Intervention phase. In Experiment 1, participants received fully random feedback, whereas in Experiment 2 participants received a mixture of random and veridical feedback.

This design closely follows Crossley et al. (2013), which showed that fully random feedback does not produce masking or overwriting of procedural category knowledge, whereas mixed feedback appeared to do so. We included both a fully random intervention (Experiment 1) and a mixed-feedback intervention (Experiment 2) because the present design differs slightly from Crossley et al. (2013) in stimulus properties and category structures. To distinguish between overwriting and masking, we presented the following on-screen message just before the first trial of the Test phase:

> Over the last many trials the feedback you received was random. This was an important part of the experiment. From now on, the feedback will again be valid. Please keep trying to categorize correctly. Press the Y key to proceed.

If mixed feedback overwrites existing procedural knowledge, participants should be forced to reacquire that knowledge from scratch, and performance in the Test phase should not exceed that of the Learn phase. By contrast, if mixed feedback only masks prior learning, participants should be able to use the instructions to re-express their procedural knowledge, leading to Test-phase performance comparable to, or better than, that observed at the end of learning.

Experiment 1, in which the intervention was fully random feedback, served as a control condition to replicate the earlier finding that such feedback does not produce overwriting. This step was important because the present study used slightly different stimuli and category structures from the earlier work.

Stimuli and categories

Visual stimuli were circular sine-wave gratings that varied in spatial frequency and orientation. To generate the stimuli, we created a set of 900 unique (x, y) coordinates uniformly distributed within the elliptical region shown in Figure 1. To accomplish this, the coordinates of all stimuli were generated by first sampling points

in polar coordinates and then converting them into Cartesian coordinates. Specifically, radius values r were sampled from a uniform distribution on the interval $[0, 1]$, and angle values θ were sampled uniformly from the interval $[0, 2\pi]$. These polar coordinates (r, θ) were then transformed into Cartesian coordinates (x, y) using the equations $x = r\cos(\theta)$ and $y = r\sin(\theta)$. This resulted in a set of (x, y) coordinates uniformly distributed within a circle of radius 1 and centered at the origin. Next, (x, y) coordinates were transformed from a circular uniform distribution to an elliptical uniform distribution with its major axis parallel to the x-axis by multiplying the x values by 124.02 and the y values by 28.44. For the Test phase in the New Learning conditions, the coordinates were rotated by 90 degrees about the centroid of the full stimulus set. The resulting stimulus distributions are shown in Figure 1. Finally, each x coordinate was transformed into a spatial frequency value and each y coordinate was transformed into an orientation value using the following equations:

$$x_t = x \cdot \frac{5}{100},$$
$$y_t = \left(y \cdot \frac{90}{100}\right) \cdot \frac{\pi}{180}$$

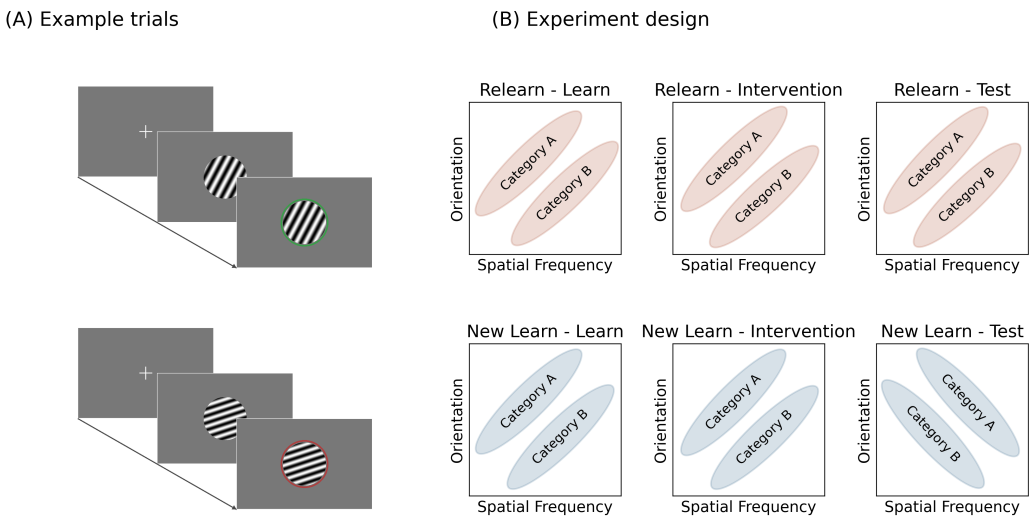

Figure 1: Stimuli and experimental design. **A:** Example trial sequences showing fixation, sine-wave grating presentation, and feedback for correct (top) and incorrect (bottom) responses. **B:** Category distributions used across experiments. In the Relearn condition, the Learn and Test phases used the same information-integration category structure. In the New Learn condition, the Learn phase used the original category structure and the Test phase used a rotated category structure.

Category learning can be described within a multiple-systems framework that distinguishes between declarative and procedural mechanisms [19, 20]. Declarative memory-based category learning relies on explicit reasoning and hypothesis testing. Rule-based (RB) tasks are commonly used to study the declarative system, as they typically require participants to apply simple, verbally describable rules (e.g., if spatial orientation exceeds some threshold angle, choose Category A; otherwise, choose Category B).

Procedural memory-based category learning, by contrast, depends on the gradual formation of stimulus–response associations acquired through reinforcement of direct experience. Information-integration (II) tasks are typically used to study the procedural system. These tasks require learners to combine information across multiple stimulus dimensions in a way that cannot be easily verbalized. Unlike RB learning, where performance often improves suddenly following discovery of the correct rule, II learning is characterized by incremental trial-by-trial improvements [21, 22].

Evidence that RB tasks are primarily learned with declarative systems and II tasks are primarily learned with procedural systems comes from a wide range of behavioral, neuroimaging, and patient studies. For reviews, see [15, 16, 17]. In the present study, we employ II categories because they specifically engage procedural mechanisms.

Procedure

Participants provided informed consent and were given an optional demographic questionnaire to complete. Participants were instructed to categorize circular sine-wave gratings on the basis of their spatial frequency and orientation, and were told that a stimulus from either category was equally likely on any given trial. Each participant completed a single session consisting of 900 trials. Each phase (Learn, Intervention, Test) consisted of 300 trials. On each trial, participants viewed a fixation cross (1000 ms), followed by a stimulus whose appearance was terminated by the participant pressing either the "d" or "k" key on a keyboard to indicate their category choice. Following their response, feedback was presented about whether their categorization was correct or incorrect (1000 ms). Correct responses were indicated by a green circular halo that appeared around the stimulus. Incorrect responses were indicated by a red circular halo.

Participants

A total of 40 participants were recruited for Experiment 1 (32 female, 8 male), with ages ranging from 18 to 32 years (M = 20.33, SD = 2.61). An additional 40 participants took part in Experiment 2 (26 female, 11 male, 2 nonbinary, and 1 preferred not to disclose), with ages ranging from 18 to 22 years (M = 19.18, SD = 1.08). Participants were pseudo-randomly assigned to either the new learning or relearning condition using a blocked allocation method, ensuring equal sample sizes ($n = 20$) in each condition. To be eligible, participants had to be at least 18 years old with normal or corrected-to-normal vision. All participants were undergraduate students at XXX and received course credit in exchange for participation. Ethics approval for this study was granted by the XXX Human Research Ethics Committee (Ref#: XXX).

Participants who were best fit by a guessing model in the final 100 trials of the Learn phase were excluded from further analyses. This criterion ensured that only participants who had acquired at least a basic level of category knowledge were included in tests of overwriting versus masking. Exclusions were as follows: Experiment 1, Relearn condition (1 participants); Experiment 1, New Learn condition (3 participants); Experiment 2, Relearn condition (4 participants); and Experiment 2, New Learn condition (2 participants).

Decision-Bound Analysis

To identify the decision strategy used by each participant, we fit decision-bound models [23, 24] to the trial-by-trial response data from the final 100 trials of the Learn phase and the first 100 trials of the Test phase separately for each participant. We examined 1-dimensional rule-based models and 2-dimensional procedural models.

The rule-based models assumed that participants placed a criterion on a single stimulus dimension and categorized stimuli according to whether the stimulus value on that dimension fell above or below that criterion. For example, a participant might attend only to spatial frequency and respond Category A when the spatial frequency was relatively high and Category B when it was relatively low. These models had two free parameters, the response criterion on the attended stimulus dimension (e.g., whether the spatial frequency was considered high or low) and the variance of perceptual and criterial noise, which reflects variability in how spatial frequency is perceived and in how consistently the criterion is applied from trial to trial.

The 2-dimensional procedural models assumed that participants used a linear decision boundary with an arbitrary slope and intercept to divide the stimulus space into two response regions. That is, participants were assumed to use a general linear classifier (GLC), and stimuli were categorized according to which side of the boundary they fell on. The GLC had three free parameters, the slope and intercept of the decision bound and the variance of perceptual and criterial noise, which captures variability in how the stimulus is

perceived and in how consistently responses are mapped onto the perceived side of the boundary. For further details on the models and the model-fitting process, see [17].

Bayesian Estimation of Procedural Reacquisition

Our central question was whether mixed feedback in Experiment 2 overwrote procedural knowledge or merely masked it. If genuine overwriting occurred, participants should fail to reacquire their previously learned procedural strategy in the Test phase. If behavioral masking occurred, then reacquisition of procedural strategies should be observed once valid feedback was restored. To evaluate this, we estimated and compared procedural strategy reacquisition probabilities across experiments and conditions using Bayesian methods.

Procedural strategy reacquisition was operationally defined in terms of participants being classified as procedural (i.e., best fit by a GLC model) both in the final 100 trials of the Learn phase and the first 100 trials of the Test phase. For each condition (Relearn vs. New Learn) and experiment (Experiment 1 vs. Experiment 2), we computed a probability distribution over θ, the probability that a participant would reacquire a procedural strategy. Because each participant either did or did not reacquire a procedural strategy, the data are naturally modeled as Binomial, with n independent observations each resulting in success (reacquisition) or failure, with probability θ. We placed a Beta$(1, 1)$ prior on θ, which is equivalent to a uniform distribution over $(0, 1)$ and encodes no prior preference for any particular reacquisition rate. The Beta distribution is the conjugate prior for the Binomial likelihood, meaning that when the prior on θ is Beta and the observed data are Binomial counts, the posterior is also Beta. The parameters of this Beta are obtained by adding the observed successes and failures to the prior parameters.

$$p(\theta \mid \text{data}) = \text{Beta}(\text{successes} + 1,\ \text{failures} + 1). \tag{1}$$

We then drew $100,000$ posterior samples for each group, and used these samples to make two complementary comparisons. First, a between-experiment comparison examined whether reacquisition was less likely following the mixed feedback intervention of Experiment 2 than following the fully random feedback intervention of Experiment 1. If genuine overwriting occurred during mixed feedback but not during random feedback, the reacquisition of procedural strategies should be reduced or absent in Experiment 2 and relatively unaffected in Experiment 1. If behavioral masking occurred, reacquisition rates should remain comparable across the two experiments. To quantify this, we computed $\Delta = \theta_{\text{Exp1}} - \theta_{\text{Exp2}}$, derived 95% credible intervals, and calculated the posterior probability that $\Delta > 0$, which would indicate greater reacquisition in Experiment 1 than Experiment 2.

Second, a within-experiment comparison focused on differences between the Relearn and New Learn conditions within Experiment 2. If mixed feedback causes genuine overwriting, participants should show no advantage for Relearn over New Learn, as previously acquired procedural knowledge would be irretrievably lost. By contrast, if mixed feedback causes masking, participants in the Relearn condition should show higher rates of procedural reacquisition than those in the New Learn condition. To quantify this, we computed $\Delta = \theta_{\text{Relearn}} - \theta_{\text{New Learn}}$, derived 95% credible intervals, and calculated the posterior probability that $\Delta > 0$, which would indicate a Relearn advantage consistent with masking.

Data Availability

Data and code are available at `https://github.com/XXX/cat_unlearn`.

Results

Figure 2 displays mean accuracy per block for each condition and experiment. In Experiment 1 (left panel), participants successfully learned the category structure during the Learn phase but ceased to express this knowledge during the Intervention phase. During the Test phase, participants in the Relearn condition rapidly returned to their previous accuracy levels, indicating that the intervention (random feedback) did not erase the original learning. In contrast, those in the New Learn condition performed significantly worse,

consistent with interference from the previously learned, conflicting stimulus-response mappings. A similar pattern was observed in Experiment 2 (right panel), suggesting that the mixed-feedback intervention failed to overwrite the initial learning.

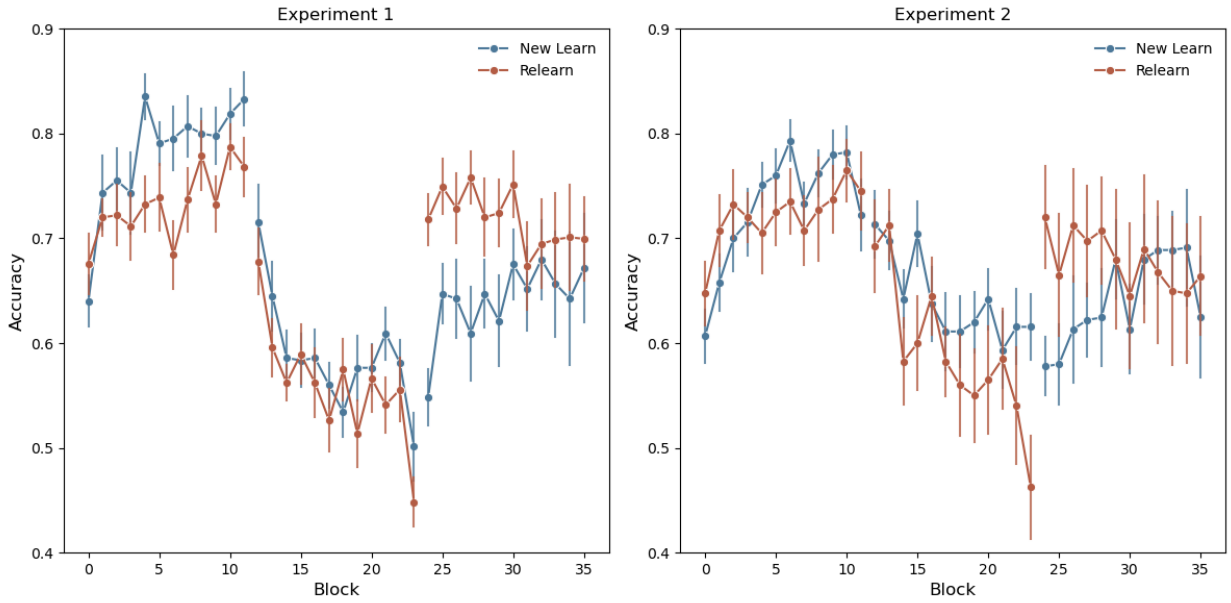

Figure 2: Accuracy over blocks excluding participants best-fit by a guessing model in the final 100 trials of the Learn phase. The Learn phase occupied Blocks 1 through 12; Intervention occupied Blocks 13 through 24; and Test occupied Blocks 25 through 36. Error bars are SEM.

Although the results suggest that the mixed-feedback intervention leads to memory masking rather than overwriting, accuracy alone cannot settle this interpretation. If procedural knowledge were truly erased, Relearn participants would be starting from scratch in the Test phase — and learners starting from scratch characteristically begin with explicit rules before transitioning to procedural output. Relearn participants' familiarity with the task could allow them to adopt such a rule more quickly than New Learn participants, producing a transient accuracy advantage that fades once New Learn participants catch up. This is consistent with the observation that the Relearn advantage in Experiment 2 spans only the first half of the Test phase, in contrast to the persistent advantage seen in Experiment 1. To rule out this possibility, we fit decision-bound models to each participant's data from the final block of the Learn phaseand the first block of the Test phase, allowing us to classify each participant's strategy as procedural or declarative. Our primary questions were: (1) did participants initially acquire a procedural strategy, and (2) did they reacquire that strategy during the Test phase? If the intervention caused true overwriting, there should be no difference in the proportion of participants expressing a procedural strategy between the Relearn and New Learn conditions during the Test phase.

Figure 3 shows heatmaps of transitions between best-fit model classes from the end of the Learn phase to the start of the Test phase, illustrating how participants shifted (or failed to shift) their categorization strategies across phases, separated by experiment and condition. In Experiment 1, participants in the Relearn condition predominantly acquired and reacquired a procedural strategy, indicating that the intervention did not disrupt their underlying category knowledge. In contrast, those in the New Learn condition tended to switch from a procedural to a rule-based strategy during the Test phase, suggesting interference rather than overwriting. A similar pattern, though less pronounced, was observed in Experiment 2. This further supports the conclusion that an intervention of mixed feedback masks rather than eliminates procedural category learning.

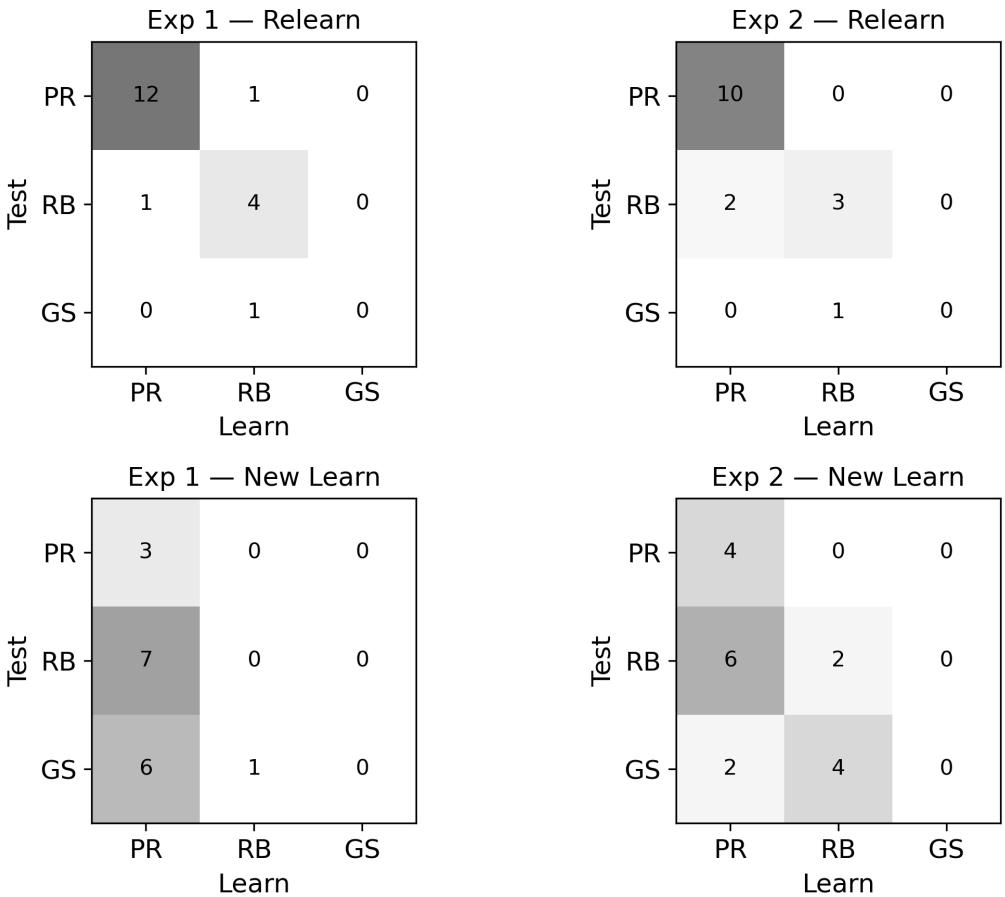

Figure 3: Heatmaps showing transitions between best-fit model classes from the end of training (x-axis) to the start of testing (y-axis), separated by experiment and condition. Each cell indicates the number of participants who transitioned between model classes. Abbreviations: PR = Procedural, RB = Rule-Based, GS = Guessing.

Figure 4 shows Bayesian posterior estimates for the probability that participants reacquired a procedural strategy during the initial Test phase trials. In Experiment 1, the posterior means and 95% credible intervals were $\hat{\theta} = 0.87$ [0.66, 0.982] for the Relearn group and $\hat{\theta} = 0.22$ [0.068, 0.433] for the New Learn group. Experiment 2 yielded similar estimates for the Relearn ($\hat{\theta} = 0.79$ [0.547, 0.949]) and New Learn ($\hat{\theta} = 0.36$ [0.138, 0.614]) groups. Between-experiment comparisons revealed little difference in reacquisition probability for either the Relearn condition ($\Delta = 0.081$ [-0.185, 0.358], $P(\Delta > 0) = 0.730$) or the New Learn condition ($\Delta = -0.135$ [-0.443, 0.170], $P(\Delta > 0) = 0.195$). By contrast, Relearn participants were substantially more likely than New Learn participants to reacquire a procedural strategy in both Experiment 1 ($\Delta = 0.644$ [0.366, 0.857], $P(\Delta > 0) > 0.999$) and Experiment 2 ($\Delta = 0.428$ [0.088, 0.720], $P(\Delta > 0) = 0.993$). Together, these results provide strong evidence that the intervention masked, but did not overwrite, procedural category knowledge.

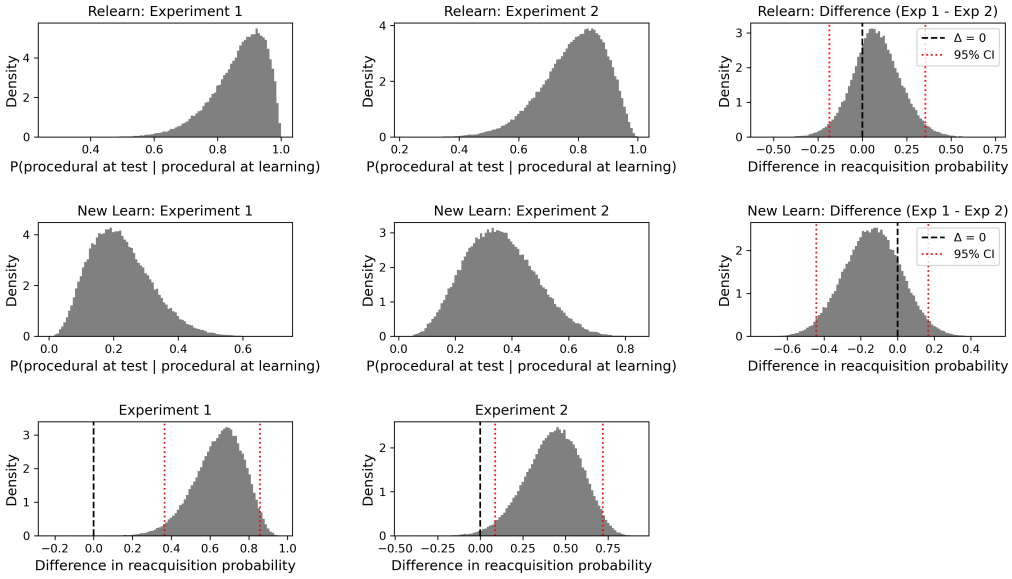

Figure 4: Bayesian posterior distributions over θ, the probability of reacquiring a procedural strategy. Top two rows show experiment comparisons within Relearn and New Learn conditions. Bottom row shows differences between conditions within each experiment. Red lines denote 95% credible intervals; black dashed lines indicate null difference ($\Delta = 0$).

Discussion

In previous work, mixed feedback – where feedback about category choices was partly random and partly veridical – appeared to induce true overwriting of procedural category knowledge. Participants who had acquired a category structure through procedural learning no longer demonstrated evidence of this knowledge after the intervention, consistent with the idea that the underlying S-R associations had been overwritten. An alternative possibility, however, is that the intervention merely masked the expression of procedural memory, leaving the underlying stimulus-response mappings intact but temporarily dormant. In the present study, we directly tested this possibility. Our findings show that mixed feedback does not cause true overwriting. Instead, it masks the behavioral expression of procedural knowledge, which remains accessible and can be re-expressed when participants are verbally cued that valid feedback has resumed.

Implications for models of procedural category learning

Ashby and Crossley [25] and Crossley, Ashby, and Maddox [18] developed a biologically grounded model of how procedural category learning is implemented in basal ganglia circuits. According to this account, category knowledge is encoded as stimulus–response associations at cortico-striatal synapses. Taken on its own, this circuit mechanism predicts that an intervention with purely random feedback should overwrite the newly formed SR maps with random associations. Our Experiment 1 results echo the earlier findings of Crossley, Ashby, and Maddox [18] and clearly show that this does not occur. Instead, random feedback leaves the original associations intact, implying that more is at play than simple reinforcement learning at cortico-striatal synapses. Specifically, there must be a mechanism that (1) detects when feedback has become random, and (2) protects initial learning from modification.

Ashby and Crossley [25] and Crossley, Ashby, and Maddox [18] hypothesized that this gating function is implemented by large aspiny cholinergic interneurons in the striatum, known as tonically active neurons (TANs) [26, 27, 28]. These neurons fire tonically in their baseline state, presynaptically inhibiting cortical

input onto striatal projection neurons [29]. When reliable rewards are present, TANs pause this inhibition, thereby permitting cortico-striatal transmission and allowing learning [30, 26, 31, 32]. When reward contingencies break down (e.g., extinction or random feedback), TANs cease pausing, shielding existing SR associations from degradation [25, 33].

This gating mechanism provides a plausible account for why learned procedural knowledge can be rapidly re-expressed during the Test phase following random feedback. Crossley, Ashby, and Maddox [18] also reported that, according to this model, an intervention composed of a mixture of random and veridical feedback should be sufficient to overwrite existing procedural category knowledge encoded in cortico-striatal synapses. In their model, the small amount of veridical feedback prevented the TANs from fully engaging their protective function because contingencies were still partially reliable, leaving cortico-striatal associations vulnerable to overwriting. Thus, the model predicted that mixed feedback should induce overwriting. This result appeared to be borne out in their empirical findings. However, the present results contradict this prediction. Mixed feedback did not erase procedural category knowledge. Rather, it simply masked its expression. This indicates that changes to the original model are required.

One potential alternative is that relevant neural circuits implement a more sophisticated mechanism that allocates, maintains, and switches between distinct context-specific sets of SR associations. This general idea has been formalized in recent mathematical models that account for a broad range of behavioral findings [34]. However, unlike our earlier work, these models are not tightly constrained by neurobiology. Recent evidence suggests that TANs may in fact support such a contextually sensitive gating mechanism. For example, these circuits are implicated in tasks that require flexible switching between behavioral policies [35]. Interpretation of these results remains difficult because the precise anatomical regions examined in much of this work tend to be associated with flexible cognitive control, and evidence for their links to SR learning remains unclear [36]. Nevertheless, some of these areas are adjacent to those highlighted in our earlier computational work [25, 18, 37, 33], which more directly implicate striatal circuits in procedural SR learning. Needless to say, further work remains to be done.

Limitations and future directions

One possible limitation of the present work concerns effective sample size. Although the raw number of participants was adequate, only a subset reliably acquired a procedural strategy during the learning phase. This necessarily reduced the number of participants available for testing our critical hypotheses. Nevertheless, our Bayesian approach allowed us to make strong inferences despite this reduction by quantifying uncertainty directly and providing posterior probabilities over key parameters.

Looking ahead, two avenues for future work are especially important. First, the computational model developed by Ashby and Crossley (2011) and Crossley et al. (2013) must be revised to account for masking rather than overwriting, so that it more accurately reflects the present findings. Second, further experimental work is needed to identify interventions capable of producing genuine overwriting of procedural knowledge. Establishing such conditions would not only clarify the boundary between masking and overwriting but also inform the design of effective strategies for modifying maladaptive habit-like behaviors.

Broader Implications

The present findings speak to a central issue in the study of procedural skills and habitual behaviors: the critical distinction between overwriting and masking. In both animal and human studies, it is well recognized that the disappearance of a behavior does not necessarily imply that the underlying associations have been overwritten. This distinction has important consequences for how we conceptualize and treat maladaptive habits.

A key question is whether effective interventions must produce true overwriting of the original stimulus–response associations, or whether masking their expression in some contexts may be sufficient. Although masking can temporarily suppress undesirable behaviors, it leaves the underlying associations intact and thus vulnerable to relapse once contextual cues or contingencies change. From this perspective, masking may be a weaker or more unstable intervention outcome compared with true overwriting. Our results therefore

underscore the importance of clearly distinguishing between interventions that induce true overwriting and those that merely mask behavior. This distinction is vital not only for advancing theoretical models of procedural learning but also for the design of practical strategies to modify maladaptive habits in real-world settings.

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